

Bidirectional Monitoring of WIPO Regulation and National Legislation: A Neuro-Symbolic Approach

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Abstract

Checking whether national copyright law actually complies with WIPO treaty obligations is still largely manual work. We built a neuro-symbolic web application that does this automatically for nine jurisdictions. A large language model handles the multilingual question, and a knowledge graph built from Akoma Ntoso XML legal documents supplies the answer. A user can ask in any language; the system finds the relevant provision, links it back to the official source, and decides Berne Convention compliance with no LLM in the loop. Beyond compliance assessment, the system provides two features aimed at practical legal research. First, every answer includes authenticated source attribution through links derived from FRBR metadata contained in the original legal documents. Second, the system supports point-in-time navigation across 68 historical versions of the Italian Copyright Act, allowing users to retrieve the version in force on a selected date. We manually validated all encoded copyright-duration values and resulting compliance classifications across the nine jurisdictions. Source code is at https://github.com/Stocastico96/ReMeP_Hackathon_26.

Keywords: neuro-symbolic AI, knowledge graph, Akoma Ntoso, copyright law, treaty compliance

1 Introduction

WIPO coordinates intellectual property law through 28 treaties spanning 194 member states, yet there is limited automated support for systematically comparing national legislation against treaty obligations. Take a question like “Is the Italian *Legge* 633/1941 compliant with Article 7 of the Berne Convention?” Answering it means a lawyer has to find the right provision, read off its number (70 years *post mortem auctoris*), and hold that against the treaty floor of 50 years pma. One country is tedious; doing it for nine by hand is slow and easy to get wrong.

LLMs can read a legal query in any language, but they also invent citations and numbers with some regularity [8]. Knowledge graphs have the opposite problem: they hold structured facts reliably, yet they cannot make sense of a question phrased in Italian or French. Our system puts each where it is strong. The LLM does the query enrichment and writes the final answer; the knowledge graph retrieves the legal evidence and runs the compliance check. The result is a **neuro-symbolic** pipeline [7] in which the hallucination-prone part never touches the numbers a verdict depends on.

This paper describes a prototype we built for the IRIS 2026 Hackathon. At its core is an end-to-end pipeline that carries a multilingual query through LLM enrichment into SPARQL retrieval over an AKN-grounded IPRonto knowledge graph, with a compliance checker that rests entirely on integer-typed RDF triples and `ipronto:compliesWith` links and keeps the LLM out of the actual decision. On top of that baseline, two features do the work that makes the output usable in a legal setting. The first is authentic source attribution: FRBR Work URIs and ELI identifiers, pulled from the AKN `FRBRWork` metadata and surfaced in the UI as verifiable links to the official legislative text. The second is point-in-time navigation, which resolves any of the 68 historical AKN expressions of the Italian copyright act (1941-2025) to whichever was in force on the date a user asks about.

2 Background and Related Work

2.1 Akoma Ntoso and the FRBR/ELI Model

Akoma Ntoso (AKN) is the OASIS/UN standard for the XML representation of legal and parliamentary documents, originally developed at the University of Bologna [9, 1]. Its hierarchical element tree (`<act>`, `<section>`, `<article>`, `<paragraph>`) assigns stable `eId` attributes to every provision, which lets us cite at the paragraph level across versions and, more importantly for a symbolic system, query a provision directly from its structural metadata without parsing any natural language.

AKN embeds a FRBR metadata layer [2] encoding the Work-Expression-Manifestation-Item hierarchy. The `<FRBRWork>` element carries: `FRBRuri` (canonical Work-level URI); `FRBRdate` (dated events typed by a `name` attribute, e.g. *dateApplicability*, *dateEntryIn-Force*); `FRBRalias` (aliases, including the European Legislation Identifier, ELI [3]); and boolean flags `FRBRauthoritative` / `FRBRprescriptive` marking official, legally binding sources. In practice these fields are populated unevenly from one national corpus to the next. For example: The UK uses `FRBRprescriptive`, Switzerland uses `FRBRauthoritative`, and Germany sets neither.

2.2 IPRonto

IPRonto is an OWL ontology for the IP domain developed by the Distributed Multimedia Applications Group (DMAG) at the Universitat Politècnica de Catalunya [4]. It defines `ExploitationRight`, `ExceptionsRight`, and properties for jurisdiction, duration, and rights grants. We use the namespace as the backbone of our KG, extending it with a `legal`: property vocabulary for SPARQL querying.

Although the current prototype uses only a limited subset of IPRonto concepts, grounding the graph in a shared legal ontology facilitates future extensions beyond copyright duration, including rights, exceptions, and licensing concepts.

2.3 Neuro-Symbolic Legal AI

LegalRuleML [5] formalises normative rules in XML with support for defeasibility and exceptions, and it gives us the formal grounding for the compliance obligations in our KG. Nardi et al. [6] survey IP ontologies elsewhere in this volume. Where we differ is in delivering a complete, runnable pipeline from LLM enrichment through to SPARQL retrieval over AKN-grounded RDF triples, with source attribution and temporal resolution on top.

3 System Architecture

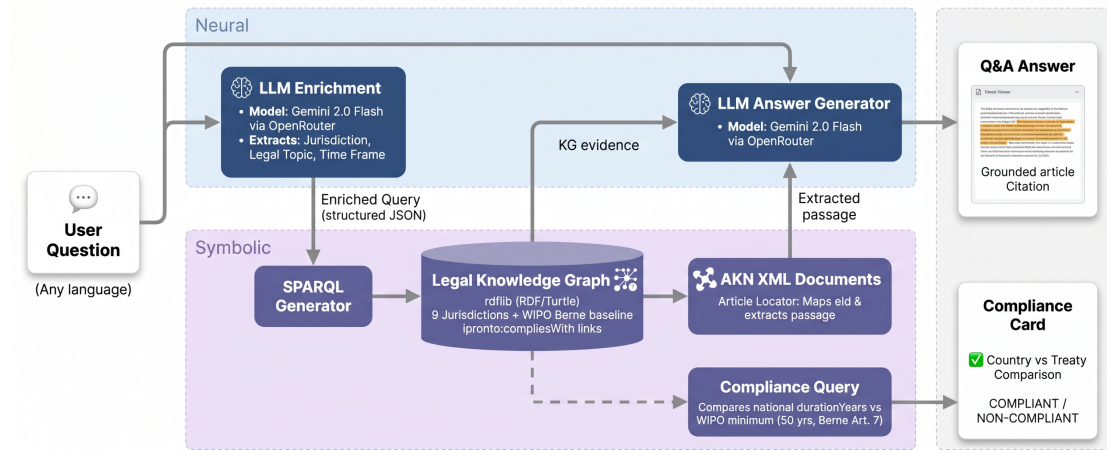


Figure 1: End-to-end pipeline. Shaded modules are neural (LLM); white modules are symbolic (SPARQL, AKN XML, FRBR). The right panel of the UI exposes the FRBR meta bar and the point-in-time date picker for Italian documents.

3.1 Stage 1: Multilingual Query Enrichment (Neural)

A user asks a question in whatever language they like. Gemini 2.0 Flash reads it and pulls out three things: the target jurisdiction, the legal topic (copyright duration, economic rights, or the personal-use exception), and a time frame. The Italian query “quando scade il diritto d’autore in Francia?”, for instance, comes back mapped to France and to copyright duration. Those three parameters are what drive the SPARQL query in Stage 2.

3.2 Stage 2: KG Retrieval (Symbolic / SPARQL)

The normalised parameters select one of three SPARQL SELECT templates executed against an in-memory rdflib graph loaded lazily on first request from a Turtle file. Figure 2 shows the knowledge graphs for Swiss and German copyright law. Each law node connects to article nodes, which in turn express economic rights.

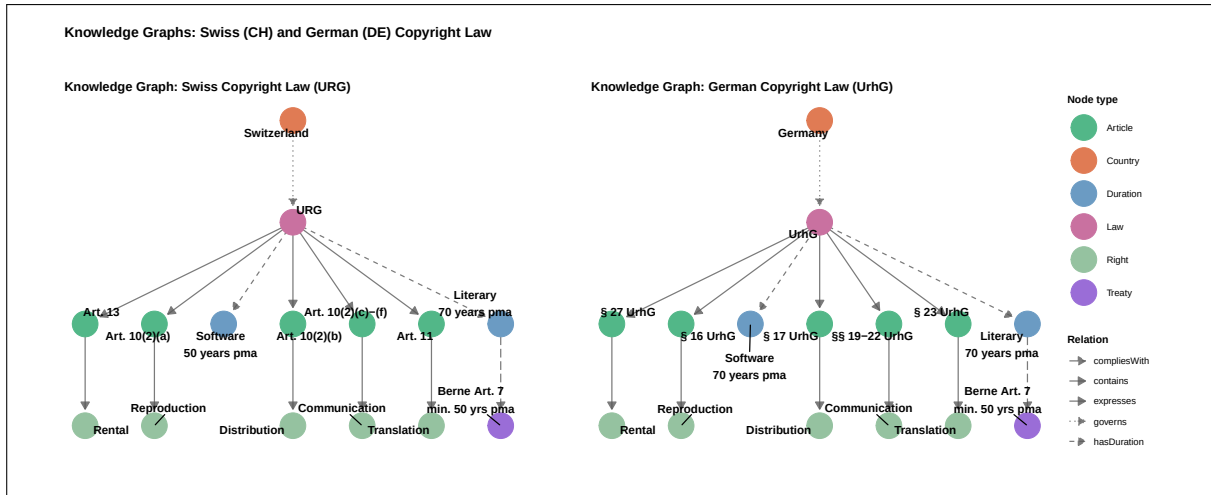


Figure 2: Knowledge graphs for Swiss and German copyright law.

We store duration values as integers rather than text, which lets the compliance checker compare them straight against the Berne minimum without any parsing. If the jurisdiction comes back unknown, the system falls back to a Berne Convention evidence record and says so to the user rather than guessing.

3.3 Stage 3: AKN Passage Extraction (Symbolic)

The article reference from the KG (e.g., “Art. 29 URG”) is used to locate the corresponding provision in the AKN XML file. The system renders the relevant article as styled HTML with the supporting text highlighted. It handles the different structural conventions used across the nine corpora. AKN files are read at query time only and are not loaded into the KG.

3.4 Stage 4: Authentic Source Attribution (Symbolic)

Every AKN document embeds FRBR metadata that pins down both the legislation and the version in force on a given date. We read four fields out of it. The Work URI is the law’s canonical identifier; in the UI it becomes a clickable “Official Source” link straight to the government portal, e.g. legislation.gov.uk for the UK CDPA. Alongside it, the applicability date records which version was in force at the queried moment, the ELI URI carries the standardised European identifier where one exists (Italy and Switzerland, in our corpus), and the authenticity flag tells the user whether the document is the official, legally binding text.

All four fields are stored in the KG at build time and returned in every API response. The document viewer shows them as colour-coded badges: green for authenticated sources, blue for the applicability date.

3.5 Stage 5: Point-in-Time Navigation (Symbolic)

Italy is the only jurisdiction for which multiple historical versions of the copyright act are available as AKN files: 68 versions of *Legge* 633/1941 from 1941 to December 2025, sourced from the Normattiva repository. The system builds an index of these versions and returns the one in force at any user-specified date. In the UI, a date picker appears for Italian documents. Selecting a date reloads the correct historical version and updates all source attribution badges accordingly.

3.6 Stage 6: Answer Generation and Compliance Checker

Stage 6 splits in two, one neural and one symbolic. For Q&A queries (6a, neural), the KG evidence, the passage, article reference, document title, and FRBR Work URI, is serialised and handed to the LLM with a grounding-focused prompt that constrains it to two to four sentences citing the article. For compliance queries (6b, symbolic), no LLM is involved at all: the checker reads the copyright duration for the queried country and the Berne minimum straight from the KG, computes the margin, and returns a structured verdict.

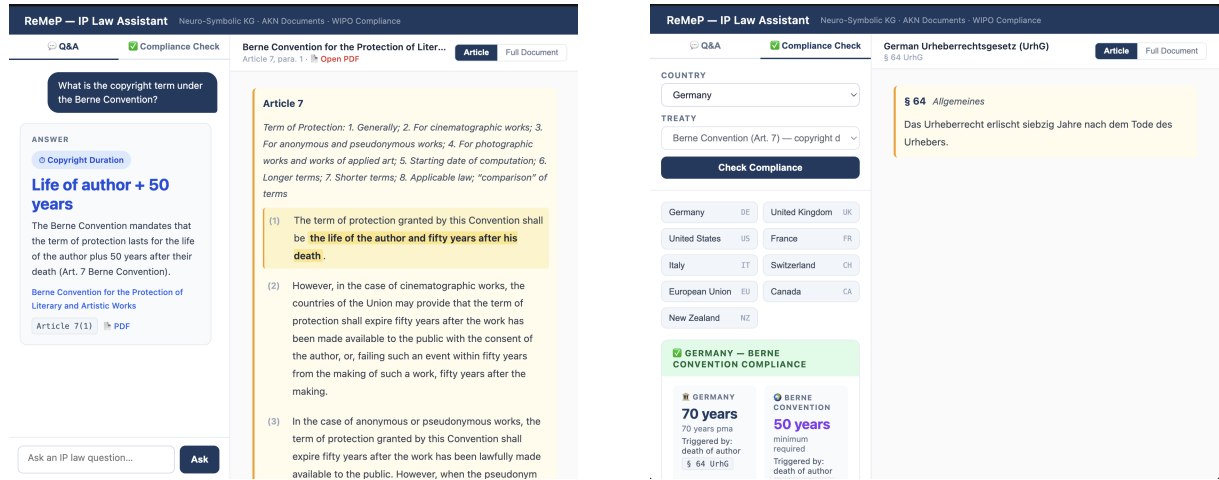
4 Implementation

The system is a Flask web application with two UI modes: Q&A and Compliance Check, accessible via tab switching. The right panel is a shared document viewer rendering AKN HTML, the FRBR meta bar, and, for Italian documents, the point-in-time date picker.

Table 1: Technology stack.

Component	Technology
Web framework	Flask (Python 3.12)
KG store	rdflib 7 (in-memory, TTL serialisation)
XML parsing	lxml
LLM	OpenRouter API (Gemini 2.0 Flash)
Schema validation	Pydantic v2
Legal documents	AKN XML (9 jurisdictions; 68 IT versions)
Ontology	IPROnto OWL [4]
FRBR extraction	<code>scripts/akn_metadata.py</code>
Temporal index	<code>scripts/point_in_time.py</code>

The KG is built on first request from a curated CSV dataset (9 rows, one per jurisdiction) and then cached in memory. FRBR metadata triples get embedded at build time through `akn_metadata`, which brings the graph to 446 triples in total.



(a) Q&A mode: answer card with citation, AKN article rendered with highlighted supporting span, and FRBR meta bar (authenticity badge, applicability date, official-source link).

(b) Compliance Check mode: side-by-side comparison of national copyright duration vs. Berne Convention minimum, with verdict and margin in years.

Figure 3: The system UI in Q&A and Compliance Check modes.

5 Evaluation

We tested the system along the same four dimensions it was built around. On the question-answering side, queries in Italian, English, French, and German are enriched correctly and routed to the right jurisdiction and rule type, and we verified the AKN passage extraction against all nine corpora, covering both `<article>` and `<section>` provision types. The grounded answers always cite the article reference that came from KG evidence, never one the LLM produced from memory.

Compliance checking illustrates the main advantage of the symbolic layer: the verdict is derived entirely from structured legal facts and explicit comparison rules rather than generated text. All nine jurisdictions come back with the correct verdict (Table 2). New Zealand, sitting at exactly 50 years, is classified as compliant with a zero margin, and every 70-year jurisdiction as compliant with twenty years to spare. The margin is computed in under 20 ms, with no LLM in the loop.

Table 2: Berne Convention Art. 7 compliance results (9 jurisdictions).

Code	Jurisdiction	Duration	Margin	Verdict
CA	Canada	70 years <i>pma</i>	+20	Compliant
EU	EU Directive	70 years <i>pma</i>	+20	Compliant
FR	France	70 years <i>pma</i>	+20	Compliant
DE	Germany	70 years <i>pma</i>	+20	Compliant
IT	Italy	70 years <i>pma</i>	+20	Compliant
NZ	New Zealand	50 years <i>pma</i>	0	Compliant
CH	Switzerland	70 years <i>pma</i>	+20	Compliant
UK	United Kingdom	70 years <i>pma</i>	+20	Compliant
US	United States	70 years <i>pma</i>	+20	Compliant

Source attribution holds up across the board: FRBR Work URIs are extracted and rendered for all nine corpora. The authenticity flags vary by jurisdiction, since the underlying metadata does: the UK marks its text `FRBRprescriptive`, Switzerland and Italy use `FRBRauthoritative`, and the rest leave it absent. ELI URIs show up only for Italy and Switzerland. Point-in-time navigation, finally, resolves the Italian versions correctly across the whole 1941–2025 span, and it does so to the day: asking for 1973-05-30 returns V5 (in force since 1955-01-27), while 1973-06-01 returns V6 (in force since 1973-05-31), one version later.

6 Limitations and Future Work

The system has clear limits, and the most interesting ones are not bugs but open questions about how far the symbolic layer can be pushed. The biggest is temporal grounding. Point-in-time navigation today works at the document level: when the user picks a date, we swap in the whole AKN expression that was in force then. The knowledge graph itself, though, knows nothing about time, since its triples carry no `valid_from` or `valid_until` markers. Adding that validity and querying it, would let the graph answer a question like “Was France compliant on 1995-01-01?” on its own, rather than by reloading a document.

The remaining limits are less about ideas and more about reach. The current graph is small by design: three rule types across nine jurisdictions, so twenty-seven core nodes, hand-curated for the hackathon. A real system would need automatic AKN-to-RDF ingestion covering a larger set of IPRonto dimensions, fed from live sources such as EUR-Lex or national gazette APIs.

7 Conclusion

The worry with putting an LLM anywhere near legal compliance work is that you trade away explainability and transparency for fluency. What this prototype shows is that you do not have to. By pairing the LLM with a knowledge graph, we keep the model where it is genuinely useful, understanding a multilingual question and phrasing an answer, while every claim that carries legal weight comes from somewhere we can point to. The compliance verdict never touches the LLM at all: it is computed symbolically, straight from the knowledge graph, so there is nothing for the model to get wrong. FRBR/ELI source attribution and point-in-time navigation over the 68-version Italian corpus do the rest, tying each answer to a specific, dated, official provision rather than to “the law” in the abstract.

The combination we find most promising is the one this prototype only begins to exploit: a knowledge graph that holds both national legislation and the international obligations it has to meet. Setting the two side by side, and letting the symbolic layer check one against the other, seems to us a direction worth pursuing well beyond copyright duration.

Source code is available at: https://github.com/Stocastico96/ReMeP_Hackathon_26

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